

Designing Your Training Program

No matter how much you climb and how hard you try, you cannot just climb your way to someday reach your genetic potential. Climbing your best requires a rigorous, nuanced training program.

Perhaps no sport can match rock climbing for the dramatic increase in the mean level of performance of its participants in recent years. Today's average climber is capable of a standard that few climbers dreamed of achieving when I began climbing in the late 1970s. Furthermore, many weekend warriors are able to progress to the lofty levels of 5.12 and 5.13—grades that hardly existed a few decades ago. The reasons for these incredible improvements include sport-climbing tactics, improved climbing shoe technology, and, more than anything else, the advent of climbing gyms and a growing focus on climbing-specific training.

Still, there is much disagreement among climbers and coaches about the best way to train, and people frequently tell me that they are confused by the often conflicting training information published in various books and on the Internet. What's more, the occasional magazine article describing the training practices of some 5.15 pro climber, while titillating, is of little use for average climbers—it might even get them injured. Other popular generalized training approaches, such as P90X and CrossFit, are similarly red herrings for a serious climber in search of the higher grades.

For a training-for-climbing workout that makes a difference, nothing beats a climbing gym with a large bouldering area, a wide variety of top rope

and lead walls, and a well-equipped training area. If there's such a facility within about a thirty-minute drive of your home, then join it! Visiting such a gym two to four days per week will enable you to engage in comprehensive training of all aspects of the climbing performance triad (skill, mind, body). Alternatively, building a home bouldering wall and training area is a wise investment if you're short on free time, due to a busy work schedule or family life, or if you live in a town without a commercial climbing gym.

Knowing what to do when you train at a climbing gym or on a home wall is what this chapter is all about. While a novice climber can steadily improve simply by going to the gym and “just climbing,” intermediate and advanced climbers must engage in a more elaborate, smartly designed program that targets known performance constraints with climbing-specific exercises. No matter how much you climb and how hard you try, you can't just climb your way to someday reaching your genetic potential. The purpose of this chapter, then, is to help you craft an intelligent, safe, and effective training program given your current fitness level and climbing ability.

First, you will learn the value of premeditating your workout and how to structure your long-term program to include microcycles, mesocycles, and an annual macrocycle—employing some form of periodization is essential to making the most of your daily workouts and obtaining maximal gains season after season. The real meat of the chapter are the sections on designing a personalized training program, including sample workouts for beginner,

intermediate, and advanced climbers. The chapter then concludes with coverage of important training considerations for youth, female, and over-fifty climbers—three classes of climbers uniquely equipped to excel in the vertical extreme!

Keys to an Effective Training Program

The premise of this book is that a well-informed, motivated, and mature individual can grow to climb at an exceedingly high level in just a few years. The preceding chapters provided a comprehensive look at the fundamental elements of climbing performance—mental, technical, and physical—and you’ve also been armed with the knowledge of an introspective self-assessment test (see chapter 2) and a basic understanding of exercise science (chapter 5). You are now in a position to execute an uncommonly effective training program—one that will help you outperform the mass of climbers who either train in a disorganized way or simply go climbing as their “training.” To be effective, the program must be designed to optimize your body composition, improve flexibility and mobility, strengthen the important stabilizer and antagonist muscles, enhance stamina and recovery ability, and

increase climbing-specific strength, power, and endurance (as shown in figure 10.1).

Premeditate Your Training

The mass of people who engage in some form of training program do so in a haphazard, ad-lib manner. There is little or no method to their madness other than to “climb a lot” and “get pumped.” This unsystematic approach will produce mediocre results and can often end in injury.

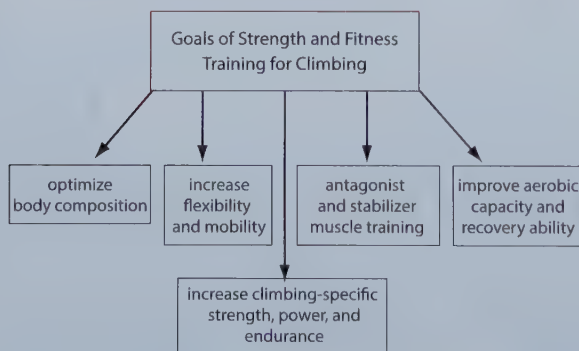
Conversely, savvy climbers are proactive in designing and modifying their training program for maximum effectiveness. Wisely, the program targets their weaknesses, is modified regularly to stave off mental or physical stagnation, and is crafted in a way to produce a peaking effect for an upcoming road trip or competition.

TARGET YOUR WEAKNESSES

Several times throughout this book, I’ve highlighted the importance of training the weakest link. For many climbers this weakest link involves poor technique, tactics, and mental control. While the ill-informed climber trains only for more physical strength, you know that it’s paramount to train your weaknesses in all aspects of the performance triad. The amount of time you dedicate to training technique, the mind, and physical strength depends on both the results of your self-assessment and your current ability level.

As a rough rule of thumb, beginner and intermediate climbers should spend a majority of their training time on improving technique, tactics, and the mental game (via climbing), while only about one-third of training time is spent doing general and sport-specific training. Advanced climbers, however, must invest much more time in the pursuit of increasing maximum strength, power, anaerobic endurance, and aerobic capacity. With already highly honed technique and mental skills, these more-veteran climbers will improve season-over-season primarily as a function of taking their physical capabilities to higher and higher levels.

Figure 10.1 Training Goals



An optimal training program will address all five areas.

REGULARLY VARY THE WORKOUT TO SUSTAIN MOTIVATION AND PHYSIOLOGICAL ADAPTATIONS

Chapter 5 explained the importance of regularly modifying your training for climbing. Sadly, many individuals go through the same basic workout ritual week after week and get frustrated with their lack of progress. Furthermore, engaging in the same weekday training or weekend at the crags will slowly quell your motivation to work hard and push beyond your current limits.

A more effective and mentally stimulating approach is to alternate your training focus every few weeks and to switch your climbing focus every few months. In the gym you can vary your training modality to target different energy systems every few workouts or weeks. Furthermore, long-term motivation, grand achievements, and the most memorable experiences in the great outdoors often come in proportion to your willingness to try new types of climbing, visit new areas, and test the limits of what is possible (given your current ability). Later in this chapter you will learn how to leverage a training mesocycle and an annual macrocycle to optimize the effectiveness of your training and help sustain improvement and psych year after year.

PRODUCE A PEAKING EFFECT FOR A ROAD TRIP OR COMPETITION

Olympic and professional athletes design their training schedules to produce a peaking phase around the time of a major event or competition. No doubt some of the best competition climbers also use peaking strategies; however, it's my observation that the majority of climbers do not deliberately plan their training in order to produce a peaking effect. Worse yet, some climbers intensify and lengthen their workouts in the days leading up to an important climb or project send day, as if they were cramming the night before an exam in school. This is a grossly counterproductive approach—when it comes to effective training, you simply can't "cram"!

Fortunately, it's really not that difficult to structure your workout schedule to peak for a road trip or to send a major project route or hardest-ever route. If you currently train and climb a few days per week, you're already doing the hardest part—the magic of peaking comes from properly tapering training volume and getting enough rest in the days leading up to the event. Details on all the above are forthcoming.

Structuring Your Workout Schedule

In this section you will learn how to manipulate your workout schedule over the time frame of several days, a few weeks or months, and a full year in order to gain optimal results. In the lexicon of sports scientists, these crucial time frames are known as the microcycle, mesocycle, and macrocycle, respectively.

MICROCYCLE

A microcycle relates to the structure, content, and volume of your training over the course of a week and for a given workout. It's essential to plan out your microcycle at least a week or two in advance—only this way can you integrate your outdoor climbing plans and your indoor training, while being sure to get enough rest days for optimal performance outside and to realize gains from your training investment. Maximizing effectiveness requires training the right things, in the right way, and in the right order during a given training session. If you've read through the preceding chapters, then you likely have a good understanding of the optimal workout hierarchy, but here's a recap:

1. Warm-up activity followed by some mobility exercises.
2. Actual climbing to develop skills and technique.
3. Performance climbing—near-maximal bouldering or roped climbing.
4. Climbing-specific exercises and energy system training.
5. Antagonist-muscle, stabilizer, core, and posterior chain training (best done on a non-climbing day).

Tapering Your Training for a Peaking Effect

For an enthusiastic, hard-training climber, nothing can be more disappointing than a poor competition performance or a "flat," weak-feeling period of outdoor climbing following a dedicated training cycle. While illness, mental stress, and poor nutrition and sleep habits sometimes contribute to lackluster performances, an inadequate end-of-cycle training taper is often to blame. If you engage in a regular three- to five-day-per-week training program, then a deliberate tapering of training activities is essential in the days leading up to a competition, road trip, or important weekend of performance climbing. Do it right and you'll be gifted with an increase in strength, power, and endurance. Conversely, a climber who tries to "cram" before an event—training more, rather than less—will likely end up feeling weak and heavy, and ultimately underperform on the rock.

The goal of tapering is to decrease accumulated fatigue from recent extensive training, while retaining hard-earned gains in strength, power, and endurance. And, if you hit the taper just right, you will likely experience a spike in maximum strength and strength/power-endurance—just what you want heading into a competition or road trip. Taper-induced adaptations that help amp up endurance include increased oxidative enzyme activity, red cell production, and supercompensation (increased intramuscular glycogen stores), whereas strength and power gains likely result from biomechanical and neural adaptations that can finally "catch up" following many weeks of intensive training.

Research shows that athletes engaged in extensive training benefit most from a one- to two-week taper. A longer tapering scheme, composed of an exponential

reduction in training volume, is best for endurance athletes logging massive mileage (Bosquet 2007), whereas a shorter (one-week) taper is generally sufficient for strength/power athletes, such as climbers. Here's how to do it.

Engage in a normal training cycle (or self-directed Daily Undulating Periodization scheme) up to one week before the event or first day of performance climbing. Then, during the final pre-performance week, progressively reduce training volume while maintaining exercise/climbing intensity. For example, if your typical climbing session lasts about three hours, then five days pre-event you want to train just two hours, and then three days pre-event you'd train just one hour. The final day or two of the taper is all about active rest and mobility work—no climbing-specific training is advised.

It's important to understand that you want to taper training volume and frequency, but not intensity. Doing a small amount of maximum-strength and power training is essential during the training taper, but you mustn't overdo it. Scale down the number of sets performed (say, on the bouldering wall, fingerboard, or campus board) by 50 percent around five days pre-event and a total of 75 percent by three days pre-event. Thus, if you normally do eight sets on the campus board, you'd cut to four sets and then two sets for the workouts five and three days before the event, respectively.

To round out your pre-performance taper, strive to nail your diet and sleep just right to help create a well-rested, light, and energetic state and, ultimately, an amazing low-gravity day of climbing!

6. Generalized aerobic training (best done in a separate workout earlier in the day or on a non-climbing day).

Determining what to do (and what not to do!) each day of the week is the real art of program design. While a climbing coach can be a big help in scheduling your microcycle and mesocycle, this chapter will equip you with the knowledge to self-coach effectively. Creating a written training game plan, at least a week or two in advance, is essential, especially if you plan to do some outdoor climbing. If you desire to be 100 percent fresh for your outdoor climbing, then you need to schedule the two prior days to be free from strenuous climbing (although a modest bout of antagonist, aerobic, and ARC training is okay two days prior).

With the possible exception of some elite climbers, a typical microcycle will include just two intensive workouts—these are the most stressful workouts composed of high-load maximum-strength and power training (weighted fingerboard hangs, campus board and such) or a high volume of strength/power-endurance (which inflicts a high acid load on the muscle cells). Other workout days will include antagonist and stabilizer training, generalized and specific aerobic training, and supplemental nonmaximal climbing. At least one day per week should allow for complete rest from training; if you climb on weekends, then Friday is typically the day of complete rest.

MESOCYCLE “PERIODIZATION”

The Principle of Variation states that you must regularly vary your workouts in order to avoid long-term training plateaus, and it’s in the mesocycle that you can best manipulate your schedule toward this end. You can leverage the proven strategy of periodization by changing your training focus, intensity, and volume every few days or weeks. Since adaptations to a specific type of workout (and training protocol) begin to flatten out after just two to four weeks, it’s essential to somehow change

your training focus and workout every few weeks. Described below are three approaches to periodizing your workouts over a mesocycle. The 4-3-2-1 and 3-2-1 cycles are best used during off-season training or in multiweek gaps between road trips, whereas the Daily Undulating Periodization (DUP) scheme is perfect for in-season use, when you are climbing outside nearly every weekend.

THE 4-3-2-1 TRAINING CYCLE

This is the training cycle that I advocate for the “average climber,” and it’s particularly appropriate if your climbing preference is roped routes. The first three phases of this cycle individually target each of the three energy systems (as shown in figure 10.2) and include a lot of actual climbing time to maximize skill development. The one-week fourth phase involves a “training taper,” composed of reduced training/climbing volume and active recovery activities. Let’s take a brief look at each phase of this cycle.

Phase 1: The four-week climbing skill and stamina phase involves, well, lots of climbing! This climbing can be done indoors, outdoors, or a combination of both. You must, however, faithfully obey an important distinction of this phase—that is, minimize time (less than 20 percent) spent maximal climbing and “projecting,” and instead log lots of mileage on a wide variety of routes that are one to three number grades below your maximum ability. The result of this four-week phase will be improved technique and tactics, acquisition of new motor programs (climbing skills), and the development of local endurance and general stamina. Climbing four days a week is ideal as long as you are not climbing at your limit or to extreme levels of fatigue. You can also engage in general conditioning exercises and stamina-building activities (specific and nonspecific aerobic training) during this phase.

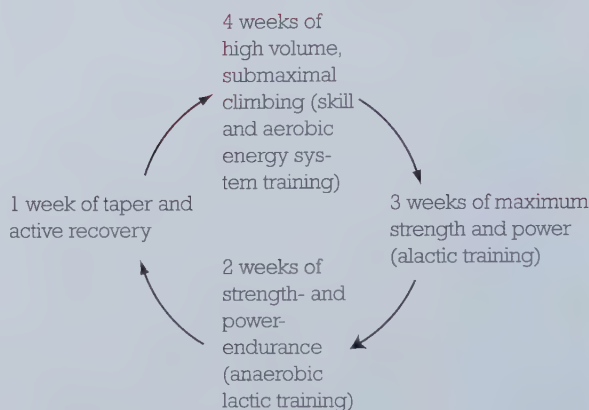
Phase 2: Three weeks of maximum-strength and power training is the next step in the cycle, and therefore hard bouldering, weighted exercises (e.g., fingerboard hangs and pull-ups), and reactive

training are ideal choices since powerful movements and short, near-maximal efforts are hallmarks of this phase. Given the high intensity and physical stress of such training, it's important to take plenty of rest between boulder problems, exercises, and workout days. Moreover, you should avoid training hard on consecutive days, although doing some moderate-intensity aerobic “recovery” climbing and antagonist training is okay on the day after a maximum-strength/power day.

Phase 3: The two-week strength/power-endurance phase is the most fatiguing and grueling portion of the cycle. Training at moderately high intensity and with reduced rest between exercises and climbing sets will produce the tell-tale muscular pump and “burn.” Interval training is the cornerstone method of triggering adaptations of the anaerobic lactic energy system. Chapters 8 and 9 present many exercises for improving strength-endurance in the finger flexors and power-endurance in the pulling muscles. I recommend three hard anaerobic endurance workouts per week with a fourth aerobic recovery climbing session. This grueling phase is best limited to just two weeks (three at most), since the anaerobic lactic energy system is the least trainable and the most stressful (most likely to lead to overtraining, excessive fatigue, and a drop in performance).

Phase 4: The final phase of the ten-week cycle is a training taper that allows for a peak in strength, power, and endurance to be revealed upon completion of the taper. Climbing and specific training must be limited to just two or three days and with a significant reduction in volume (see the “Tapering Your Training” sidebar). A small amount of high-intensity exercise is essential, so a brief (twenty-minute or less) bouldering session or two, a few sets on the hang board or campus board, and a couple of pump runs on a short route should be spread through the first five days of the week. The sixth and seventh days must allow for complete rest from climbing—only modest mobility work is advised.

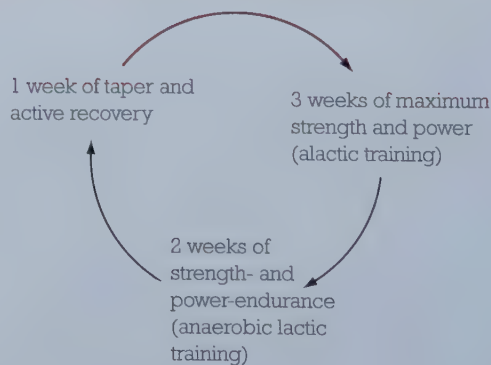
Figure 10.2 The 4-3-2-1 Training Cycle



THE 3-2-1 TRAINING CYCLE

This training cycle is ideal for climbers with a preference for hard bouldering, as it deemphasizes aerobic energy system training by eliminating the first phase of the 4-3-2-1 cycle. This can also be used during the climbing on-season, given a six-week gap between significant weekend climbing trips. As shown in figure 10.3, the 3-2-1 cycle follows the same workout strategy as the aforementioned 4-3-2-1 cycle, except for the elimination of the four weeks of skill and stamina training. Still, one day of aerobic climbing (threshold or ARC) is recommended each week

Figure 10.3 The 3-2-1 Training Cycle



DAILY UNDULATING PERIODIZATION (DUP)

This nontraditional form of periodization is a good microcycle strategy for more-advanced individuals engaging in regular outdoor performance climbing. Weekend warriors and pro climbers alike can employ DUP to help maintain all three energy systems during a period of several weeks up to a couple of months, although this approach is unlikely to yield any significant long-term gains (which requires a more focused cycle such as the 4-3-2-1 or 3-2-1).

Here's how to do it: Each one-week DUP cycle must include one or two training or climbing sessions targeting each of the three energy systems. Therefore, if you go bouldering outdoors two days per week, it's these two sessions that serve as your two maximum-strength/power workouts for the week—the remainder of your week's training

should include only one aerobic climbing session and one pumpy anaerobic capacity session. Conversely, if you engaged in two days of hard, pumpy route climbing over the weekend (two days of strength/power-endurance training), then your weekday training should involve only a maximum-strength/power (bouldering) session and a threshold or ARC aerobic climbing session. The bottom line: Regardless of your climbing preference, the goal of DUP is to engage in at least one hard workout of each energy system each week. Of course, you must also incorporate two brief sessions per week of core, antagonist, and stabilizer training—consider the latter to be mandatory.

MACROCYCLE

The macrocycle is your annual game plan of off-season training, on-season training and climbing,

Table 10.1 Sample DUP Microcycles for Boulderers and Route Climbers

Sample DUP Microcycle for Boulderers						
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
S/AA*	Aerobic Climbing & CE (AeEnd) & TC	Anaerobic Climbing (4x4s, etc.) & CE (S/P End) & TC	S/AA*	R	Hard Bouldering Outside	Hard Bouldering Outside
KEY: C=Climbing; CE (S/P End)=Strength/Power-EnduranceClimbing Exercises; CE (AeEnd)=Aerobic Endurance Climbing Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)						

Sample DUP Microcycle for Route Climbers						
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
S/AA*	Aerobic Climbing, CE (AeEnd) & TC	Bouldering & TC/CE (Str/ Power)	S/AA*	R	Hard Route Climbing	Hard Route Climbing
KEY: C=Climbing; CE (Str/Power)=Strength & Power Climbing Exercises; CE (AeEnd)=Aerobic Endurance Climbing Exercises; S=Stabilizer & Antagonist training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)						

and break periods away from training/climbing. In traditional sports the macrocycle is planned around the competitive calendar, with the goal of peaking for a major competition or sustaining near-peak conditioning throughout a sports season. In climbing, however, the goal is to structure a training program to produce a peaking effect for a major road trip, a competition, or the best climbing season in your region.

You can loosely map the macrocycle on a calendar by identifying the months of your on-season road trips or competitions, the months you expect to perform off-season training, and any time off from climbing you plan to take during the year. Note that indulging in a multiweek annual break (or brief seasonal breaks) from climbing is essential for recharging your motivation and healing any known (or unknown) injuries that may have developed during the course of the climbing season. For many climbers, taking off the latter part of December makes the most sense, as this offers the reward of holiday parties and relaxing with family after a

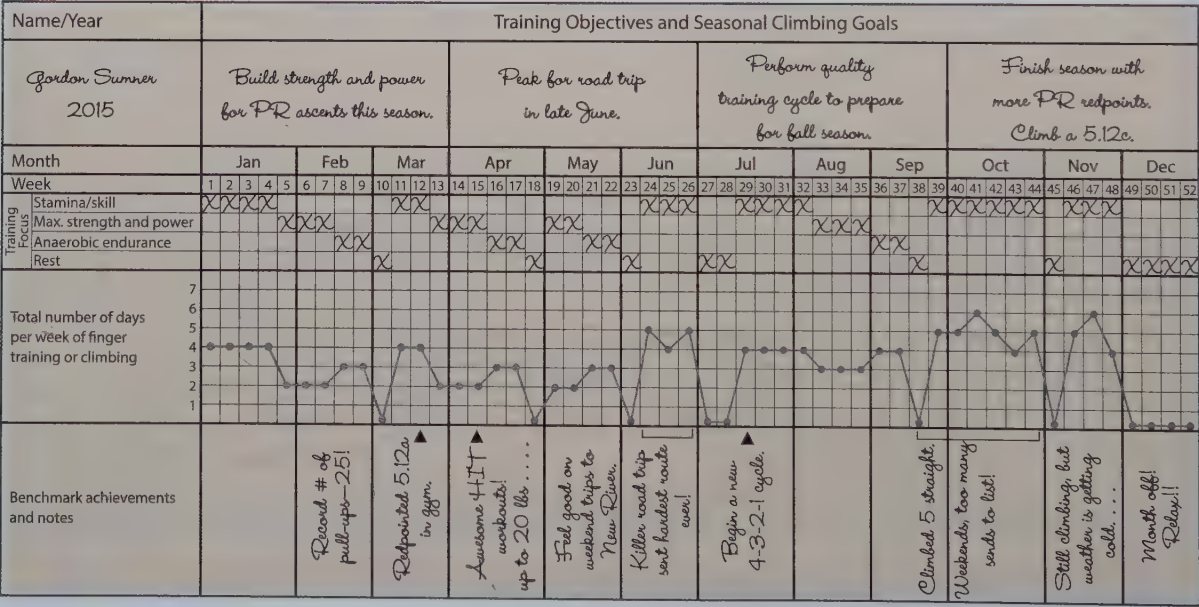
year of dedicated training and tight dietary surveillance. Another popular option is taking a couple weeks off during the peak of midsummer heat, before beginning a late-summer training cycle that leads into prime autumn climbing season.

Figure 10.4 depicts a typical macrocycle. Note that a couple of 4-3-2-1 training cycles are fit into the off-season training period, along with another during the midseason. The fall months are targeted as the peaking period, followed by a couple of weeks off to conclude the year. A blank fifty-two-week macrocycle is contained in appendix B for your use—you can also download a blank macrocycle spreadsheet at www.Training4Climbing.com.

Targeting Training on Your Preferred Subdiscipline

In chapter 1 you learned the importance of obeying the SAID Principle (specific adaptation to imposed demands) in optimizing your training for your favorite subdiscipline of climbing. Figure 1.7 depicts how the demands of these subdisciplines vary over

Figure 10.4 Sample Macrocycle



a continuum from bouldering to alpine climbing. Maximizing the effectiveness of your training requires designing your workouts accordingly.

Most climbers reading this book participate in the three subdisciplines of bouldering, sport climbing, and multipitch climbing, and this text is obviously focused on helping these climbers improve their performance. Still, big-wall and alpine climbers should be able to glean plenty of useful information. For instance, in accordance with the SAID Principle, a serious alpine climber would benefit much more from high-volume endurance StairMaster training and trail running than from bouldering on a home wall or hanging on a fingerboard. Of course, the most specific and effective training for alpine climbers is to engage in frequent, high-volume climbing and nonspecific aerobic training.

Conversely, building a home wall or joining a climbing gym is the single biggest advantage that boulder, sport, or multipitch climbers could give themselves. Time spent training in the gym should mimic the performance demands of their preferred focus. Boulders must dedicate more time to maximum-strength and power training, while multipitch and big-wall climbers should focus more on climbing-specific aerobic training and generalized endurance exercise. Sport climbers possessing a high degree of technical skill, however, would do best to cycle their focus between maximum strength/power, anaerobic capacity, and climbing-specific aerobic capacity.

Clearly, the best training program for you will change over time as your technical ability and your physical strengths and weaknesses change. For this reason active self-coaching, with regular self-assessments and course correction, is critical in maintaining a successful training program. The time invested in plotting your program intelligently and striving to stay on course over the long term will pay huge dividends in how far and how fast you progress in this sport. You might also consider hiring a climbing coach to help guide your training—the

objective analysis and expert guidance can be a real wild card in obtaining the most rapid gains in ability possible.

IMPORTANCE OF A HOME WALL OR GYM MEMBERSHIP

Regardless of your ability, nothing beats indoor climbing for sport-specific, time-efficient training, any time of the day or year. Hopefully, there is a good commercial facility within a reasonable distance of your home or workplace. If so, join the gym and use it at least twice per week—this is the number one thing you can do to improve climbing ability and fitness. Many of us are less fortunate (the climbing mega gym nearest to my home is more than an hour away), however, so it's vital to invest in a home wall or some basic exercise equipment (fingerboard, TRX trainer, dumbbells) in place of that gym membership.

If your space is tight, simply build an eight-foot by eight-foot, 50-degree overhanging wall with a small section of ceiling climbing atop it. While this setup has obvious limitations (physical and mental), it will enable you to get an excellent upper-body workout as well as help improve some aspects of climbing movement and body position. If a larger space is available, it would be wise to construct three additional sections of wall: a less overhung wall (20 to 30 degrees past vertical is ideal, especially if you want to make a System Wall), a super-steep, 65-degree overhanging wall, and a slightly overhanging (about 5 degrees past vertical) traverse wall. A garage with a high ceiling offers a good location for your home wall, especially if there is a way to control the climate in the summer and winter; however, many homeowners feel it's more practical to build their home gym in the basement or spare room.

Another excellent strategy is organizing a community wall. Recruit five or ten energetic climbers to pitch in a few hundred dollars each. Rent a garage or some similar structure that has room

to build several hundred square feet of climbing surface with a variety of angles, and then complete your facility with a campus board, pull-up bar, gymnastic rings, HIT System, Treadwall, and fingerboard.

Designing an Effective Personal Training Program

A powerful central tenet of this book is that you must train in the manner best suited to you and not do as others do. Obviously, there is a limit to how precisely I can prescribe an optimal training program for you via the static format of a book. On the pages that follow, however, I will lay out the basic, fundamentally important guidelines that you should follow in developing your own training

program. Do so and you will be far better off than the typical climber who trains in a haphazard, trial-and-error manner.

Most important, you must adopt the appropriate training template for your current ability level. Over time you can tweak this program according to the results you experience and in line with the good training sense you've developed from reading this book. As long as you act in ways consistent with the principles and concepts described throughout this text, you will remain on course toward your goals. The starting point of your program design will be one of the three workout templates—beginner, accomplished, and advanced—detailed on the pages that follow. Consult table 10.2 to determine which classification best describes your current climbing ability and experience.

Table 10.2 Climber Classifications and Training Program Overview

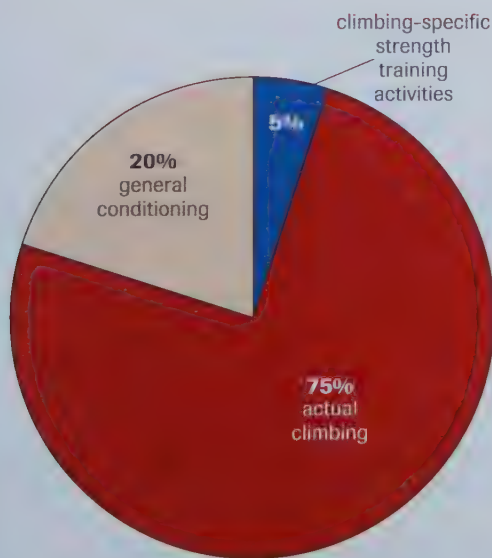
	Beginner	Accomplished	Advanced
Classification of Ability	• Less than one year climbing experience • Toprope ability 5.10 and under • Little or no lead-climbing • Boulder V3 and under	• Actively climbing more than one year • Toprope 5.11 to 5.12+ • Lead climbing up to 5.10+ trad or 5.12 sport • Boulder V4 to V8	• More than three years' climbing experience • On-sight lead ability of 5.12 sport or 5.11+ trad • Redpoint climbing 5.13 and above • Boulder V9 or harder
Training Program Overview	• Climb: 2–4x per week • General conditioning and stability/antagonist training: 2x per week • Generalized aerobic conditioning: 1–4x per week	• Climbing and climbing-specific exercises: 3–4x per week • Stability/antagonist training: 2x per week • Generalized aerobic conditioning: 1–3x per week	• Climbing and climbing-specific exercises: 3–5x per week • Stability/antagonist training: 2x per week • Generalized aerobic conditioning: 1–2x per week

Beginner-Level Workout

PRIMARY MISSION

Learn climbing skills and techniques for efficient movement, optimize body composition, and improve general conditioning, including exercises that target the antagonist and stabilizer muscles. Time spent climbing will provide a rigorous climbing-specific (muscle and tendon) workout, and thus little or no additional climbing-specific training is needed. Also of high importance is a conscious quest to develop basic mental skills such as hold/move identification, mental rehearsal and sequencing, visualization, arousal control, fear management, and goal setting.

Figure 10.5 Workout Time—Beginner



Beginners should invest the majority of their training time in actual climbing. Time spent fitness training should mostly focus on general conditioning.

WORKOUT GUIDELINES

Climb up to four days per week, including outdoor climbing days whenever possible. Reduce body fat, if needed, via improved dietary surveillance and up to four days of aerobic training per week. Train the antagonist, stabilizer, and core muscles twice per week using the exercises described in chapters 6 and 7. If you cannot climb during the work-week, you should instead engage in some of the entry-level finger flexor and pull-muscle exercises described in chapters 8 and 9. Time spent training and climbing should break down according to figure 10.5. An overview of beginner workouts, with and without climbing wall access, is shown in table 10.3. Visit www.Training4Climbing.com for detailed workouts available as free downloads.

SAMPLE MICROCYCLES

Two slightly different microcycles are shown in table 10.4—one for those able to climb during the workweek (home or commercial gym, or outdoors) and the other for those unable to climb except on weekends.

SAMPLE MESOCYCLE

I do not advise the use of a formal training mesocycle (such as the 4-3-2-1 cycle) for a true novice, nor do I suggest you attempt to climb at maximum difficulty or push excessively hard on the rock. Instead, your medium-term goals should be to increase your volume of climbing as well as the diversity of techniques used. If you climb regularly, it's prudent to take a few consecutive days off from climbing every few weeks to allow for systemic consolidation of skills and strengthening of tendons (which take much longer than muscles to strengthen).

SAMPLE MACROCYCLE

A thoroughly designed and executed macrocycle is less important for beginners than it is for

Table 10.3 Beginner Workout Guidelines

Beginner’s Workout with Climbing

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 20–40 minutes submaximal practice climbing—focus on learning new skills, efficient movement, and mental skills.
- 10–30 minutes strenuous bouldering or near-limit roped climbing.
- 3–5 sets pull-muscle exercises (see table 9.1).
- 15–30 minutes of core, antagonist, and stabilizer muscle training, doing an aggregate of 8–12 sets; although this can be done on a separate day.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Visit www.Training4Climbing.com for additional workouts as free downloads.

Beginner’s Workout with No Climbing

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 5–8 sets pull-muscle exercises (see table 9.1).
- 4–6 total sets of various core exercises.
- 2–4 sets of fingerboard repeaters.
- 15–30 minutes of antagonist and stabilizer muscle training, doing an aggregate of 8–12 sets.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Table 10.4 Beginner Sample Microcycles

Four-Day-Per-Week Climbing

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or S/AA*	C/CE/TC	S/AA*	C/CE/TC	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE=Climbing-Specific Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

Weekend-Only Climbing

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or AA	CE/S/TC	R or AA	CE/S/TC	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE=Climbing-Specific Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

accomplished and elite climbers, though you can still benefit from some long-term planning relating to possible road trips and multiweek breaks from

climbing. Still, you can use the blank macrocycle in appendix B to plot your training and climbing schedule as it evolves.

Summary of Training for a Beginning Climber

- Climb up to four days per week—and outdoors as often as possible. Climb for volume over difficulty—maximum learning of a wide range of skills and tactics is far more important than the grade of routes sent.
- Engage in conditioning exercises that focus on optimizing body composition, improving flexibility, and strengthening the important antagonist, stabilizer, and core muscles. Improved dietary surveillance is crucial for improving body composition.
- Climbing-specific strength training should be limited to the beginner exercises (tables 8.1 and 9.1), and actual climbing should be generally given preference over doing these exercises.
- Strive for awareness of your thoughts and emotions throughout the day and while climbing. Practice the mental techniques described in chapter 3, and begin using them as part of your preclimb preparations and while on the rock.

Accomplished Climber Workout

PRIMARY MISSION

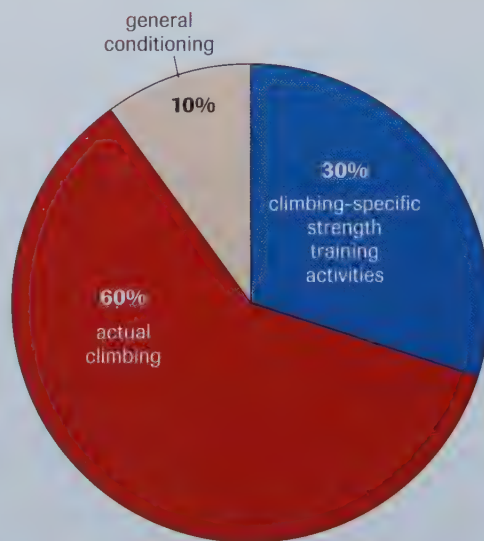
Maximize economy of climbing movement with constant focus on refining mental and technical skills. Expand technical skills and on-sight climbing ability with experience on many different rock types and by avoiding specialization as long as possible. Increase intensity of climbing-specific training exercises with twice-weekly finger flexor training on a fingerboard and a similar amount of focused pull-muscle and core training—getting stronger year-over-year must be the prime directive of your training-for-climbing program. Concurrent stabilizer and antagonist training is a must to maintain muscle balance and prevent injury.

WORKOUT GUIDELINES

Every intermediate-level climber has a different “perfect” training program depending on his or her unique physical, technical, and mental weaknesses, as well as time available and climbing goals. Still, actual climbing is the most essential part of this workout program for all accomplished climbers. Climbing three to four days per week (indoors and/or outdoors) is the backbone of this program. For most people, this will require joining a climbing gym or building a home training wall to help facilitate weekday climbing. Strive to climb outside as often as possible—one or two days each weekend, if possible—to enable learning of the diverse skills and techniques not represented by human-made walls.

Serious strength training can be scheduled via the DUP scheme and, when feasible, according to the 4-3-2-1 or 3-2-1 training cycles.

Figure 10.6 Workout Time—Accomplished



Accomplished climbers should spend about one-third of total training time on climbing-specific exercises, but time spent climbing rock and plastic remains the top priority.

Table 10.5 Accomplished Climber Workout Guidelines

Sample “Skill and Stamina” (Aerobic Energy System) Workout (2–4 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 75–120 minutes practice climbing and lapping moderate (submaximal, no-fall) routes. Build up to 1,000 feet of climbing; if the gym's routes average 30 feet in length, you'll need to climb 30+ routes (or up and down 15+ times).
- 10–30 minutes strenuous bouldering or near-limit roped climbing.
- 15–30 minutes of antagonist and stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do this on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Sample “Maximum-Strength/Power” Workout (1.25–2.5 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 30–60 minutes maximal bouldering and hypergravity bouldering.
- 20–40 minutes maximum-strength and power-training exercises. Select appropriate forearm exercises (3–7 total sets) and pull-muscle exercises (3–7 total sets) from tables 8.1 and 9.1, respectively.
- Conclude with 3–6 sets of core exercises.
- 20–40 minutes of posterior chain and antagonist/stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do these exercises on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Sample “Strength/Power-Endurance” Workout (1.5–2.5 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 30–60 minutes interval training on rope routes or a Treadwall, or bouldering 4x4s or similar. You should get deeply pumped, yet strive to avoid complete muscle failure until late in the session.
- 20–40 minutes strength/power-endurance exercises. Select appropriate finger flexor exercises (5–10 total sets) and pull-muscle exercises (5–10 total sets) from tables 8.1 and 9.1.
- Conclude with 3–6 sets of core exercises.
- 20–40 minutes of posterior chain and antagonist/stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do these exercises on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Visit www.Training4Climbing.com for additional workouts as free downloads.

Actual climbing should be done before doing any climbing-specific exercises, and never climb or do climbing-specific exercises more than a total of four days per week. Train the antagonist, stabilizer, and core muscles twice per week, on either climbing or nonclimbing days (ideal). Maintain close dietary surveillance during the latter portion of off-season training and during on-season training/climbing. Your time spent training and climbing should break down in the proportions shown in figure 10.6.

Use the three sample workouts in table 10.5 as templates from which you can build the optimal workout for you, or visit Training4Climbing.com for detailed workouts available as free downloads. Modify and vary the program over the meso-cycle and macrocycle and also in accordance with the principles of training detailed in chapter 5. In the context of the 4-3-2-1 training cycle, use the stamina and skill template for four weeks, the maximum-strength and power for three weeks, and the strength/power-endurance for two weeks.

SAMPLE MICROCYCLES

Your discipline at planning and executing quality workouts over a seven-day microcycle is a primary factor in determining the results you obtain. Integrating climbing days, hard training days, and sufficient rest days is the crux of the matter. Train too little or rest too little, and you will shortchange yourself. Table 10.6 provides four microcycles to help guide your scheduling. Depending on your access to a climbing wall or rock, select either the weekend-only climbing microcycle or the four-days-climbing microcycle. Use the microcycle as a template for planning your weekly schedule, but recognize that you may need to adjust or deviate from the schedule depending on travel or the need for additional rest.

MESOCYCLE DESIGN

Dedicated periods of serious strength training for gains are best scheduled according to the 4-3-2-1 cycle. Figure 10.2 shows a highly effective ten-week program that allows for adequate rest and thus produces maximum-strength gains with low risk of overtraining or injury. High-end accomplished climbers may benefit more by using the 3-2-1 mesocycle (see figure 10.3) or DUP scheme.

Peak climbing season (when you most often climb for performance) will make it difficult to follow a strict ten-week mesocycle. At these times you may deviate from the schedule by taking two full rest days before a weekend climbing trip and at least one active rest day after the weekend climbing. Often the best in-season training strategy is to develop a seven-day microcycle that best accommodates your weekend climbing while still allowing for at least one “maintenance” climbing-specific strength-training workout and one stabilizer and total core session—in doing this you’re more or less engaging in a DUP program. Regardless, when you find a weekly schedule that works, stick with it for several weeks or until you can commit to a more structured 3-2-1 or 4-3-2-1 regimen.

SAMPLE MACROCYCLE

The average accomplished climber gets outdoors twenty or more weekends per year and may go on as many as two to four extended road trips per year. Therefore, careful macrocycle planning is vital to optimize preparation for these trips and to help produce peaking for an extreme project or personal-best ascent. Use the blank macrocycle in appendix B to fashion an effective long-term training plan that accounts for your travel plans, the best outdoor climbing season, and when you choose to take your month off from climbing.

Table 10.6 Accomplished Climber Sample Microcycles

Four-Day-Per-Week Climbing—Schedule #1

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or S/AA*	C/CE/TC	S/AA*	C/CE/TC	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE=Climbing-Specific Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

Four-Day-Per-Week Climbing—Schedule #2

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or S/AA*	C/CE/TC (Str/Power)	C/S/CE (AeEnd)	TC/AA*	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE (Str/Power)=Strength & Power Climbing Exercises; CE (AeEnd)=Aerobic Endurance Climbing Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

Weekend-Only Climbing—Schedule #1

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or AA	CE/S/TC	R or AA	CE/S/TC	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE=Climbing-Specific Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

Weekend-Only Climbing—Schedule #2

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
R or S/AA	CE (Str/ Power)/TC	CE (AeEnd)/S	TC/AA*	R	C/AA*	C/CE/TC*

KEY: C=Climbing; CE (Str/Power)=Strength & Power Climbing Exercises; CE (AeEnd)=Aerobic Endurance Climbing Exercises; S=Stabilizer & Antagonist Training; TC=Total Core & Leg Exercises; AA=Nonspecific Aerobic Activity; R=Rest (*Optional)

Summary of Training for an Accomplished Climber

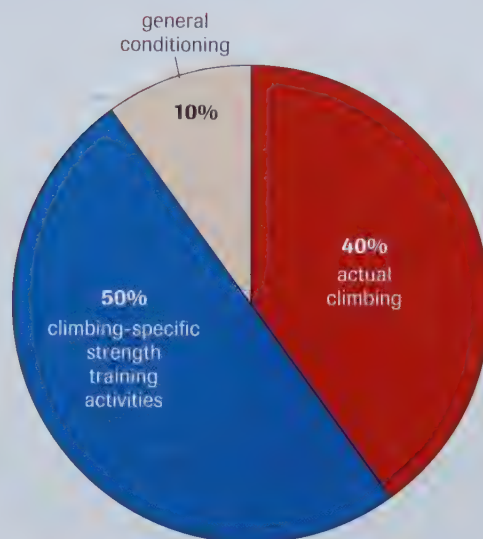
- Climb up to four days per week and gain exposure to as many different types and styles of climbing as possible. Refine mental and technical skills to maximize economy of movement—the fastest way to becoming a better, stronger climber.
- Engage in regular, scheduled, climbing-specific strength training to increase maximum grip strength, upper-body power, and endurance. Focus primarily on the intermediate exercises listed in tables 8.1 and 9.1. During a particular workout, always perform actual climbing before engaging in strength-training exercises.
- Commit to training the antagonist and stabilizer muscles twice per week—critical for maintaining muscle balance and lowering injury risk.
- Work on becoming a more mental climber, on a mission to always climb and train smartly. Plan your workouts and then work your plan—follow-through is critical to long-term improvement. Practice using mental-training strategies throughout the week, and then strive to leverage all your mental and tactical techniques when you step onto the rock.

Advanced Climber Workout

PRIMARY MISSION

Identify and correct any technical weak spots or energy leaks (no matter how small) that compromise climbing performance. Constantly evaluate and refine mental skills—there's always room for improvement in this area, even among top pros! Work to eliminate subtle forms of self-sabotage by narrowing your focus onto the process of climbing and letting go of any outcome-oriented thinking.

Figure 10.7 Workout Time—Advanced



Only advanced climbers need to focus extensively on climbing-specific strength-training activities—getting stronger, season-over-season, is priority number one. Design workouts to target a preferred subdiscipline, a specific energy system, or simulate a current project route.

Exercise workouts must be highly specific to your preferred climbing subdiscipline (bouldering, sport climbing, multipitch, or big walls), and they must stretch the limits of what you are currently capable of doing, in order to eke out further neuromuscular gains and central governor recalibration. Sound nutrition and sleep habits are critical for accelerating recovery and obtaining optimal adaptations from your hard workouts.

WORKOUT GUIDELINES

Unlike the mass of climbers, the elite performer needs to spend a disproportionate amount of training time on strength training (see figure 10.7). For boulderers and sport climbers, several weeks or even a month or two at a time may be dedicated to strength/power training and hard bouldering

Table 10.7 Advanced Climber Workout Guidelines

Maximum-Strength/Power Workout (1.5–2.5 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 30–60 minutes maximal bouldering and hypergravity bouldering.
- 30–45 minutes maximum-strength and power-training exercises. Select appropriate forearm exercises (7–10 total sets) and pull-muscle exercises (7–10 total sets) from tables 8.1 and 9.1, respectively.
- Conclude with 4–8 sets of total core.
- 20–40 minutes of posterior chain and antagonist/stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do these exercises on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Strength/Power-Endurance Workout (1.5–2.5 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 45–60 minutes interval training on near-limit rope routes or a Treadwall, or bouldering 4x4s or similar. You should get deeply pumped, yet strive to avoid complete muscle failure until late in the session.
- 30–45 minutes strength/power-endurance exercises. Select appropriate finger flexor exercises (7–10 total sets) and pull-muscle exercises (7–10 total sets) from tables 8.1 and 9.1.
- Conclude with 3–6 sets of core exercises.
- 20–40 minutes of posterior chain and antagonist/stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do these exercises on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Aerobic Capacity Workout (2–4 hours)

- 10–15 minutes of general warm-up activity including a few mobility exercises.
- 90–180 minutes of intermittent climbing (submaximal, no-fall) routes. Build up to 1,200 feet of climbing; if the gym's routes average 30 feet in length, you'll need to climb 40+ routes (or up and down 20+ times). Alternatively, climb large holds on a Treadwall for 3–5-minute intervals at an intensity of between 5 and 8.5 out of 10. Do up to 8 intervals with approximately 1:1 climb-to-rest ratio.
- 3–6 sets of core exercises.
- 15–30 minutes of antagonist and stabilizer muscles training, doing an aggregate of 8–12 sets—ideally, do this on a rest day from climbing.
- 5–15 minutes cool-down—foam rolling and mild stretching.

Visit www.Training4Climbing.com for additional workouts as free downloads.

with little or no actual roped climbing. Multipitch and big-wall climbers must invest a large amount of time into actual climbing—climbing for volume, exercising to build strength/power-endurance, and stamina-training activities are what it's all about. Frequent training with the advanced-level exercises in chapters 8 and 9 is essential to achieving further physical gains. Elites with several years of strength and power training under their belt must also train to develop higher aerobic power and aerobic capacity—essential for reaching your genetic potential for sustained, near-limit ascents.

Access to an indoor climbing wall, boulders, or crags is absolutely necessary for advanced-level training. Use of the 3-2-1 cycle can be highly effective for both on- and off-season training, although many advanced climbers will train intuitively via autoregulation and the DUP scheme. Still, the microcycles, mesocycles, and macrocycle described herein offer a framework for such intuitive training. Finally, brief but regular bouts of stabilizer, antagonist, and posterior-chain training will lower injury risk and help optimize climbing posture for maximal application of force to the rock. Abundant rest and proactive performance nutrition are vital to accelerate recovery and help stave off injury and overtraining.

Table 10.7 provides three highly focused workout templates, one each for training maximum strength and power, strength/power-endurance, and climbing-specific aerobic capacity. You will need to modify the volume of training as well as exercise selection based on your capabilities, time available to train, and your climbing preference. Design and vary your workouts according to the training principles outlined in chapter 5.

SAMPLE MICROCYCLES

The appropriate amount of training stimulus (workout length, frequency, and intensity) will vary greatly among elite climbers based on climbing

preference, quality of nutrition, genetic factors, and age, among other things. Some individuals may do best by training hard just twice per week, while others do better by training and climbing a total of four days per week. Given the tendency of elite climbers to push their limits, however, it is often necessary to take forced rest days instead of training or projecting while sore. As a guiding principle, advanced climbers should “train hard when training, and rest completely when resting.” While many elites will choose to train intuitively via DUP, you can also adopt one of the accomplished climber weekly programs detailed in table 10.6.

SAMPLE MESOCYCLE

As in planning weekly training and climbing, many advanced climbers will intuit their multiweek mesocycle. Even so, it's critical that you consider your outdoor climbing and travel plans, then build a workout schedule around these dates. One planning strategy is to schedule rest and workout days backward from your climbing days—for example, allow for at least forty-eight hours of rest before a weekend of performance climbing and then plan a workout day or two prior to these rest days. On a calendar, mark in your climbing days and trips, then mark the preceding rest days, then fill in your training days. Advanced climbers preferring a more structured schedule will do best by planning their workouts according to the 3-2-1 mesocycle shown in figure 10.3.

SAMPLE MACROCYCLE

Advanced climbers tend to be very goal-oriented and compulsive long-term planners. It's in the macrocycle that you need to book long road trips, competitions, and an annual month off from climbing, so that you can plan the most effective training schedule to produce a physical peaking for key events. You can plot your own macrocycle using the blank chart found in appendix B.

Summary of Training for an Advanced Climber

- Plan and execute an optimal—not maximum—strength-training program that targets your personal weakest link. The goal is to increase grip strength, upper-body power, and anaerobic endurance season over season.
- Constantly evaluate your technique and mental performance to identify subtle flaws that are preventing further gains. Economy of climbing movement is paramount, and it's your mental, technical, and tactical skills that determine your fuel efficiency. Make the mental strategies described in chapter 3 into life skills that you employ daily.
- Err on the side of over-resting instead of over-training. Use performance nutrition and generous amounts of sleep and rest to enhance recovery and maximize gains from training.
- Be a compulsive planner of training, travel, and rest. Try to leave nothing to chance, and avoid trial-and-error training or getting drawn into some else's (flawed) training routine.
- Evaluate all you do in your daily life with this question: *Is it helping me reach my goals or holding me back in some way?*

Training Considerations for Female, Over-Fifty, and Youth Climbers

This final section takes a look at special considerations for female climbers, those over age fifty, and youth climbers. I've intentionally kept this section brief, because I feel that these groups are particularly gifted for climbing—there are only a few things they need to consider or do differently compared with others.

Unique Issues for Female Climbers

Female climbers differ from males both physiologically and psychologically. While the psychological

differences are more difficult to measure, there are clear physiological differences that may be an asset or liability in terms of climbing performance. The average female is about five inches shorter, thirty-five pounds lighter in total body mass, and forty-five pounds lighter in lean body mass (due to a higher percentage of body fat) than the average male. This large difference in lean body mass (muscle) is largely attributed to greater levels of the hormone testosterone in males (Bloomfield 1994).

Consequently, the average female possesses approximately 40 percent of the upper-body strength and 70 percent of the lower-body strength when compared with men. In terms of strength-to-lean-body-mass, however, the ratio is notably less—females possess about 55 percent of the upper-body strength and about the same in lower-body strength as men (Wilmore 1974). Thus, it's clear that the greater level of adipose tissue in females has a negative effect on physical performance, especially in sports requiring a high strength-to-weight ratio. The female climber, therefore, will benefit much more from regular aerobic training (to lower her percentage of body fat) than her male counterpart—running for twenty to forty minutes a few days per week will effectively increase strength-to-weight ratio in many females (due to changes in body composition).

Though females are naturally weaker than males (less testosterone and lean mass), they do respond to strength training in the same ways as men. Consequently, the serious female climber should not hesitate to engage in the climbing-specific strength-training exercises described in chapters 8 and 9. In particular, the focus should be on increasing maximum strength in the finger flexor and pull muscles and general conditioning of the core, antagonist, and stabilizer muscles.

The first few weeks or months of training will produce marked improvement thanks to neural adaptations. Beyond that, strength gains will come more slowly as hypertrophy (muscle growth) becomes a more significant player in producing

strength gains. For this reason it's vital that strength training for the female climber be just as focused as for the male. The typical health club workout is no more appropriate for female climbers than for males, and in the long run could have a negative effect on climbing performance.

Technically and mentally, the beginning female climber is often a better performer than her male counterpart, and this is something that every female climber should recognize and leverage to the greatest possible extent. More flexibility, a lower center of mass, less body weight, shorter fingers, and a more measured approach to climbing can all help a female climber outperform her male counterpart despite possessing less strength. So while sport-specific strength training is a must for any serious female climber, she should not overlook her gifts of style, strategy, creativity, and finesse.

As a final note, it's a common misconception that physical performance tends to be worse during menstruation. While the menstrual cycle's effect on performance varies widely among individuals and from sport to sport, at least one study has shown that grip strength was greater during the actual menstrual phase (Davis 1991). Ultimately, you need to develop an awareness of just when is your best performance time of the month. You can then plan your training and climbing to exploit this period, whether it's for a few high-intensity workouts or making a personal-best ascent!

Conditioning for Over-Fifty Climbers

As adult climbers age, numerous physiological changes combine to form an increasing constraint on performance, especially beyond the age of fifty. A few of the unfortunate changes include reduced VO_2 max (aerobic capacity), decreased muscle mass, a lower proportion of fast-twitch muscle fibers, and reduced recovery. Despite these inevitable life changes, you can still climb at a very high level given a renewed focus on the mental and technical aspects of climbing and a steady dose of strength training. I know of more than a few fifty- and

sixty-somethings who regularly climb 5.12 and 5.13 (at age fifty-two, I'm now one of them—yikes!), ascend big walls, and trek in the mountains. You can too, given a three-pronged approach of injury avoidance, physical conditioning, and mastery of skills.

INJURY AVOIDANCE

Compared with the resilient bodies of young adults, older climbers' muscles, tendons, and joints are more susceptible to injury during rigorous workouts and near-limit climbs. Common issues range from muscle pulls to dislocated shoulders, torn tendons, and a variety of other joint and spinal problems. Fortunately, you can significantly reduce your risk by engaging in a comprehensive warm-up before every training and climbing session. Younger climbers might rush through a warm-up in just a few minutes, but older climbers would be wise to complete a full thirty-minute warm-up of general aerobic activity, light exercise, stretching, mobility activities, and easy climbing. Such a progressive warm-up will markedly decrease injury risk by warming and lengthening the muscles and spreading synovial fluid to lubricate the joints. While thirty minutes of non-climbing exercise might not be your idea of a good time, it will enhance the quality of your climbing and reduce the risk of muscle or joint injury that might lay you up for months or even knock you out of climbing completely.

Another way the mature, disciplined climber can avoid injuries is simply by avoiding potentially injurious moves while climbing. The goal is to foster a level of kinesthetic awareness at which you can assess—or often intuit—the risk potential of a given move. Whether it is an awkward-feeling drop-knee, a tweaky-feeling pocket, or an improbable-feeling lunge, your discipline to heed the sensory feedback and rapidly evaluate the situation before forging onward can save you. Ultimately, you will need to make a quick decision as to whether you should retreat from the risky-feeling move, test the move once to see how it feels, or just push onward in the belief that you will succeed without incident. As a

rule the older you get, the more you should view such a risky-feeling move as a stop sign instead of a caution sign.

PHYSICAL CONDITIONING

Physical conditioning for over-fifty climbers is not all that different from the program I prescribe for the mass of climbers. You can safely employ many of the exercises contained in this book. Most of your limitations relate to dynamic, forceful training exercises, which become increasingly dangerous with advancing age. Climbers over fifty years of age would be wise to not engage in the most dynamic forms of campus training, one-arm lock-offs or one-arm pull-ups, frequent lunging, and V-hard bouldering. Of course, every climber possesses different genetic encoding, experience, and physical capabilities, so there are surely a few senior climbers who prevail through the most stressful endeavors. But for the vast majority of older climbers, dynamic training is dangerous training. Otherwise, your fitness-training goals are similar to those of every other climber: optimize body composition, improve aerobic capacity and stamina, and increase muscular strength and endurance.

Preplanning workout and rest days is of great importance for the older climber. Too many back-to-back workout (or climbing) days, too little rest, and poor nutrition over just a few consecutive days will crack open the door to possible injury or illness. Compound this over several weeks and it will fling the door wide. Once an older climber is injured and sick, reduced immune efficiency and changing hormone levels mean slower recovery and a faster drop-off of physical conditioning than for a younger climber. The bottom line for over-fifty climbers: Train, rest, and eat on a calculated schedule that will reduce injury risk, and do nothing to tempt injury.

TECHNICAL AND MENTAL MASTERY

The best older climbers are usually Zen masters who leverage the fact that climbing performance is two-thirds technical and mental and only one-third

physical. By exploiting superior skill and wisdom, and bringing many years of experience to the table, older climbers can become true masters of rock by climbing very near their maximum capacity. Whether that top capability is 5.8 or 5.14, you can spot these elder masters by their measured approach, smooth sailing through scary terrain, and even the occasional calculated lunge or grunt that shows they are still willing to pull out all the stops to send.

Developing such mastery takes many years; in fact, in a complex sport such as climbing you can still acquire and refine mental and technical skills after ten, twenty, or even thirty years' experience. So while your physical capacities may be steady or waning, you can often compensate by improving mentally and technically. Consequently, you should strive to strike a balance between strength training—still an important part of the equation—and going climbing at one of the myriad wonderful crags around the world. And, after all, isn't that the bottom line? Simply by moving over stone, you tap into the life force that climbing provides, which transcends ability, gender, and age. That's the power of climbing!

Training for Junior Climbers

Kids can unquestionably learn complex sports skills more rapidly than adults, and in recent years we've seen numerous "wonder kids" take the sport-climbing world by storm. Clearly these young, generally prepubescent climbers possess the slight physique ideal for difficult climbing. There are numerous other physiological traits that work in their favor to enhance function and rate of recovery. Unfortunately, these young climbers often lack the maturity, self-awareness, and life experience to transfer their sport-climbing prowess to a wide range of climbing pursuits. They are also not prepared for the rigors of serious climbing-specific training as outlined in this text. Let's take a look at the appropriate type of activity and training for youth climbers.

With the growing popularity of youth climbing competitions and the recent press of preteen



SKILL DEVELOPMENT

Climbing is first and foremost a skill sport, and it should always be treated as such, especially when dealing with youth climbers. Much like golf, climbing requires proficient use of a wide range of skills, while strength and power are secondary. Consequently, skill development is paramount for novice climbers (of all ages), while extensive strength training is unnecessary and inappropriate early on. Interestingly, preteen brains are predisposed to prolific learning of motor skills, and so they have a time-limited gift to develop and refine climbing skills at a much higher rate and with greater ease than adults.

While climbing in general is extremely intuitive—and many youngsters might seem to be able to successfully teach themselves—there are in fact many complex techniques and skills that novices are unlikely to discover on their own. This is where knowledgeable climbing coaches are indispensable, since they can present new skills to be learned with proper form and in a logical progression, all the while keeping the climbing activities safe and fun.

A good coach will have a lesson plan of sorts that begins with a warm-up activity and then progresses through several periods focused on learning new skills, practicing previously learned skills, engaging in some form of “performance climbing” (attempting to climb entire roped routes or “limit” boulder problems), and concluding with some basic conditioning exercises and cool-down stretching. All the while the coach will engage the students, actively correcting lackluster form and inefficient movement patterns.

Feedback is essential to the learning process—the coach must help direct effective practice that will encode fundamentally correct movement skills into the brain and foster a high level of climbing efficiency, rather than allowing youth climbers to muscle through a move with poor technique and call it “okay.” As in learning any physical skill, it’s paramount that young climbers avoid encoding sloppy, inefficient ways of moving that will be more difficult to correct once they are firmly wired in the brain.

Ashima Shiraishi climbed 5.15a before she was old enough to drive! CLAUDIA ZEIGLER

climbers sending V12 and 5.14 routes, many parents—and some coaches as well—are suspicious that these elite youth climbers must be involved in some secret, arduous strength-training program. The truth is, however, that elite youth climbers develop as the result of an early introduction to climbing (often between ages four and seven), high-quality coaching, dedicated practice, and a slight (often pre-pubescent) physique.

So what is the best way for a youth climber to train, and what can a coach or parent do to foster a youngster’s interest in climbing, while at the same time keeping it safe and fun?

As both a parent and coach of two elite youth climbers, I’ve spent much time the last decade researching and developing a youth climbing philosophy that I feel is safe, effective, and appropriate. On the pages that follow I’ll present—for coaches and parents—this philosophy for training youth climbers, including guidelines for training skill and strategy, cognitive development, physical fitness training, and injury prevention.

(Again, golf is a good analogy: Many adult golfers futilely struggle to overcome technical flaws that were deeply encoded during their formative days.)

Longer term, youth climbers should be encouraged to explore unfamiliar terrain. An early focus on slab climbing should expand to the more challenging terrain of overhanging walls and bouldering caves. When they've achieved a high level of proficiency at face climbing, youths should be exposed to cracks and corners, which demand acquisition of new skill sets. An early focus on bouldering and top rope climbing should eventually progress to an introduction to lead climbing, although this is obviously a risky, headier form of climbing that is not to be rushed into prematurely. Some strong, talented youth climbers will develop the skill and confidence to begin lead climbing by age eight or nine, while others may take years longer. There are no rules (no specific age) for when a climber is ready to lead—obviously this is a critical decision for the coach and parent to make together, and not something the youth climber should unilaterally decide.

A final and most important coaching matter involves climbing frequency and degree of dedication to the sport. Understandably, many youths fall in love with climbing—to the point that they would like to make it their one and only recreational/sporting activity. It's my opinion, however, that single-sport specialization should be discouraged before the age of thirteen and, ideally, until after age sixteen. Research has identified a finite “golden period” of accelerated acquisition of motor abilities that lasts only through the early teenage years. This period of heightened neurodevelopment should be used to learn a wide range of sports skills that include running, jumping, flipping, throwing and/or shooting a ball, swinging a club or racquet, and swimming, among many other wonderful and pleasurable ways to engage in skilled movement. A youth's brief opportunity to learn at hyperspeed should be invested in learning more than just how to dance and pull up a rock wall. It's my experience that young athletes can come to climb at a

national-class level (or higher) while at the same time participating in—and perhaps excelling at—one or two other sports, and in school, as well!

COGNITIVE DEVELOPMENT

My advice on cognitive training for youth climbers parallels almost everything I've outlined above on training technical skills. Start with the most basic mental challenges—developing confidence and poise in climbing hand-over-hand up juggy vertical routes—and gradually introduce more challenging terrain that requires visualization of sequences and the vertical chess match of reading several moves ahead while on a climb. For some kid climbers the process is a natural and apparently simple one to learn, as they quickly solve climbing wall puzzles on the fly. Other kids will require more coaching input to help them solve the puzzle and “see” the moves to be performed. (A laser pointer is a must-have coaching tool for helping a young climber see and mentally rehearse a tricky sequence.)

For younger climbers—say, under age nine—it's important to treat a climbing session as more a playground experience than a training session. A good coach will present climbing games that keep the focus on having fun, with learning of climbing skills and strategies a by-product.

Climbers between the ages of nine and thirteen can be presented with more complex instruction on critical cognitive skills such as mental rehearsal, visualization, arousal and fear control, risk management, and such. With good instruction, it's within this age range that many youth climbers, usually those with several years of experience, put together the complex technical and mental puzzle that is hard climbing and suddenly break into the lofty grades of bouldering V8 and/or lead climbing 5.12 (or harder).

Given a growing mastery of the fundamental technical and mental skills, some youth climbers between ages eight and thirteen are ready to be introduced to the challenges and joys of outdoor climbing. As with the introduction to lead climbing

indoors, however, taking youths outside is a big step that should not be rushed. The earliest transitioners to outdoor climbing will be those youths whose parents are experienced climbers themselves—this way the parent can be responsible for making the tough calls on what climbs are safe or unsafe for a youth to attempt. For those kids without a climbing parent, it's best to wait until at least thirteen to venture outdoors—and even then, to do so only in the presence of a mature, qualified climbing coach and/or instructor.

No matter if the outdoor climbing adventure is bouldering, top roping, or lead climbing, the game is pretty much the same as it is indoors, although the technical and mental demands are far greater on the infinite playing field of Mother Nature's rock walls. Given a good coach or experienced parent, however, climbing outdoors offers an extraordinarily rich environment for learning and experiencing the wonders of nature. The bottom line: Outdoor climbing can be nearly as safe as indoor; however, some outdoor climbs are more complex, and this may increase the potential for climber and belayer error, in addition to the common objective dangers that come into play in the great outdoors. Proceed with caution.

Just as outdoor climbing will stretch a youth climber's cognition in new and exciting ways, so will competition climbing—which, for many youths who don't venture into outdoor climbing, offers an excellent way to add a more challenging cognitive aspect to the game. Many climbing gyms hold local youth competitions, which serve as a great introduction to the unique challenges and pressures of competition climbing. More formalized regional competitions for both bouldering and roped climbing are professionally organized and sanctioned by the governing body USA Climbing. Youths between the ages of eight and eighteen compete in five different age divisions for medals and to qualify for a national championship event. Learn more at USAClimbing.org.

AGE-APPROPRIATE STRENGTH TRAINING

The increasingly widespread push for early specialization in youth sports has led to an unnecessary and, in some cases, inappropriate application of advanced-strength-training techniques among young athletes. Regardless of the sport—soccer, baseball, running, or rock climbing—early specialization and use of adult-like training programs has sadly led to a rapid increase in overuse injuries. It's now well documented that some of the hardest training/climbing youths, both in America and Europe, are suffering growth plate fractures in the fingers—a few top climbers have had multiple fractures, some leading to permanent disfiguration.

It's interesting to note, however, the countless youth climbers who “climb their age” without engaging in dangerous, age-inappropriate training practices. While some form of physical training is certainly necessary, there's really no need for an intensive strength/power-training program—the risk is far greater than the reward. By simply climbing three or four days per week for several years, many youth climbers quickly advance into the higher grades of sport climbing—the wild card seems to be easy access to a climbing gym and high-quality coaching.

Given a regular schedule of climbing, a limited amount of supplemental strength and conditioning exercises is okay and, likely, beneficial for building a complete, well-balanced athlete. Disregard any old wives' tales you've heard that “strength training is not for kids” or that “strength training will stunt a youth's growth.” The American Academy of Pediatrics and the American College of Sports Medicine both support basic strength training for kids, as long as the exercises are age appropriate and not overly dynamic or stressful, and youths are coached to exercise with proper form. So what are the age-appropriate exercises for youth climbers? Let's take a look at three general classifications: ages six through nine, ten through fifteen, and sixteen through eighteen.

AGES SIX THROUGH NINE

Prior to the adolescent growth spurt, most apparent gains in strength come from motor learning, not hypertrophy (muscle growth), and so any manner of extensive strength training is unnecessary and largely a waste of time at this age. Motor learning comes when adaptations of the nervous system—specifically, improved coordination and synchronization of muscular motor units (groups of muscle fibers that fire together)—lead to a greater expression of strength and power. Therefore, movement-oriented training should be the backbone of every workout. Developing climbing-specific strength, then, is simply a matter of following through with a consistent schedule of climbing (which will yield measurable gains in strength despite little or no change in muscle size).

Use of a few supplemental climbing-like exercises is fine, as long as it doesn't take away from movement training and climbing-for-fun time. Body weight exercises such as pull-ups, push-ups, various core training exercises, and other similar gymnastic movements are the only strength-training exercises needed at this age.

AGES TEN THROUGH FIFTEEN

Given a year or more of climbing experience and assuming a solid command of technical skills, it is not inappropriate to introduce a moderate amount of strength training for climbers ages ten through fifteen. These are the years of the adolescent growth spurt, with peak height velocity occurring at age 11.5 for girls and age 13.5 for boys. Changing hormone levels will lead to a noticeable growth of the muscles as weight velocity peaks between ages twelve and fifteen.

It's during this period that some fast-growing youths perceive a disturbing decrease in strength-to-weight ratio as their weight gain outpaces their strength gains. Once again, quality coaching is critical so that concerned youth climbers understand why they are apparently getting weaker despite a consistent climbing schedule. Some misguided

youths will react by beginning a strict diet or extensive strength- or power-training program (or both) in an attempt to regain their physical capabilities. This response is unfortunate and, in fact, unhealthy, and it often ends in injury, anorexia, or burnout.

A good coach will explain the changes that all youth climbers face during puberty, and how the end result will undoubtedly be a physique that is stronger, taller, more powerful, and supremely capable of climbing at a very high level. The coach should direct a gradual introduction of general and climbing-specific strength-training exercises that will train the newly gained muscle to be more efficient and effective for performing in the vertical plane, as well as reduce injury risk as the growing youths attempt harder climbs.

With the overriding training emphasis remaining on the development of outstanding climbing technique and mental fortitude, youth climbers can engage in twenty to forty minutes of strength training, three days per week. Climbing-specific exercises such as pull-ups, lock-offs, open-hand-grip fingerboard hangs on medium-size holds (in small doses), slow, controlled campus laddering (never double dynoing!), and various core-strengthening routines should be executed on climbing days, not rest days. Most important, the youth should forgo most advanced and stressful exercises (as used by older elite climbers), such as double-dyno campus training and weighted fingerboard hangs—these can lead to growth plate fractures in the fingers and a necessary withdrawal from climbing for an extended period.

Perhaps the most important addition to a youth's training program is a small number of exercises to target the muscles that oppose the prime movers in climbing. A regular schedule of climbing (even without use of climbing-specific strength exercises) will lead to significant gains in strength of the pull muscles that are the prime movers for the vertical athlete. Maintaining stable joints and proper posture, reducing injury risk, and pursuing peak performance in climbing demands a steadfast commitment to training the antagonist push muscles

of the upper body. Specifically, the training should target the extension-producing muscles of the chest, shoulders, and arms, the scapular stabilizers, and the small muscles of the rotator cuff. A modest investment—twenty to thirty minutes, two or three days per week—will strengthen these important yet often overlooked muscles, and vastly lower the risk of the elbow and shoulder injuries common among avid and expert climbers alike.

Most of the necessary climbing-specific and antagonist-muscle-training exercises are described earlier in this book. For the antagonist muscles I suggest beginning with two sets of push-ups and reverse wrist curls, TRX trainer Ys and Ts, one or two sets of dumbbell shoulder presses, two sets of dips (with assistance, if needed), and two sets each of external and internal rotations of the shoulder. Each exercise should be performed with a relatively light weight that allows fifteen to twenty repetitions. These exercises can be performed at the end of a climbing workout or on rest days from climbing.

A final facet of effective training for youth climbers is flexibility training. While most climbing moves do not require extraordinary flexibility, quickly growing bones and muscles can lead to an increasing sense of tightness that may somewhat limit movement on the rock. A moderate amount of daily stretching can go a long way toward maintaining smooth, flexible movement. Gentle stretching of the arms, torso, and legs should be performed as part of a more comprehensive warm-up before engaging in maximal climbing. Furthermore, engaging in ten to fifteen minutes of stretching each evening (while watching TV or during quiet time before bed) is highly effective in helping relax and lengthen the tired, growing muscles of a hard-training youth climber.

AGES SIXTEEN THROUGH EIGHTEEN

As youth climbers approach their adult height, they can begin a gradual transition into a more intensive climbing-specific training program. Still, growth plates remain vulnerable to injury; they may not fully fuse until around ages eighteen to twenty.

What's more, increases in training volume and magnitude can lead to overuse and acute injuries of the fingers, elbows, and shoulders. For these reasons, among others, a fully engaged coach or climbing parent is highly beneficial during this transition into becoming astute, self-directed adult climbers.

Many of these late-teen climbers will already be highly accomplished, and a few will be proven to possess world-class capabilities. Elite-level training techniques, such as one-armed pull-ups, weighted fingerboard hangs, campus training, and weighted bouldering (say, with a ten-pound belt), can be added incrementally during this multiyear ramp-up period. However, hold off on fully dynamic double-dyno campus training and advanced hypergravity-training protocols until after the eighteenth birthday (and even later in the case of a late bloomer). Strengthening the antagonist muscle groups remains essential for muscle balance, proper posture, and reducing the risk of elbow and shoulder injuries—all potential issues for hard-training late-teen and twenty-something climbers.

All the while, it's imperative that the training program remain focused on improving the complete climber—including mental and technical training—rather than a narrow, impulsive quest for greater strength and power as the sole pathway to improvement. Teenage climbers should be reminded frequently that sharp advances in ability often result from mental breakthroughs, and, furthermore, that becoming a true master of rock comes only by way of gaining vast experience on a wide range of climbing walls and rock types. Toward this end, a good coach will constantly dovetail physical training with ongoing instruction on advanced climbing techniques, tactics, and the many facets of the mental game. The most thoughtful and introspective teenagers will be ready to begin a study of the book *Maximum Climbing: Mental Training for Peak Performance and Optimal Experience*—a challenging text that explores powerful metacognitive skills that will enhance performance in climbing as well as many other life endeavors.

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